

■ PROJECT PLAN — EDUMIND AI CHEMISTRY TUTOR

Timeline: 13 weeks (12 dev + 2 QA)

Architecture: 5 Features, prioritized for MVP delivery

MONTH 1 — FOUNDATION + CORE AI

FEATURE 1: AUTH + CURRICULUM ENGINE

Week 1: Planning

- Defining Features For the MVP
- Meeting with Team to discuss about the market and competitors.
- Meeting with Team to get to know each other experiences
- ■Assign a Features to Each person

Week 2: Infrastructure + Design System

- Docker Compose (PostgreSQL + Redis + nginx + API)
- Alembic migration setup
- Flutter: Riverpod + GoRouter + flutter_localizations
- Design system from prototype (mock prototype, AppColors, AppTypography, AppTheme)
- Dio API client with JWT interceptor - init APIS
- Flutter: Splash screen + 3-screen Arabic onboarding
- Flutter: Arabic ↔ English language toggle

Week 3: Auth + User Model

- Backend: User model (name, email, grade, interests, teaching_style)
- Backend: Register + Login + JWT endpoints
- Backend: Quiz + QuizAttempt models
- Flutter: Login + Register screens (RTL, validation)
- Flutter: Token persistence + auto-login - Cache
- TTS voice pronunciation for chemistry terms , Voice input (Whisper transcription)
- Flutter: Dark/Light theme toggle

Week 4: Curriculum + Navigation

- Backend: Chapter + Lesson + LessonProgress models
- Backend: Seed Grade 9 chemistry data (4 chapters, 18 lessons)
- Backend: [GET /chapters](#), [POST /lessons/{id}/complete](#)
- Backend: AI quiz generator (MCQ, T/F, fill-in-blank per lesson)

- Submit + instant correction + personalized explanations
- Weak topic detection logic
- Flutter: Bottom nav shell (Home, Lessons, Lab, Profile)
- Flutter: Lessons screen (ChapterCard, LessonRow, progress bars)

MONTH 2

FEATURE 2: AI TUTOR CHAT (RAG)

Week 5: RAG Pipeline + Chat Backend

- Backend: pgvector extension setup on PostgreSQL
- Backend: Textbook PDF → chunk → embed pipeline (sentence-transformers)
- Backend: ChatSession + ChatMessage models / history
- Flutter: Chat screen (RTL bubbles, markdown, typing dots)
- Flutter: Streaming text display
- Flutter: Teaching Style switcher + format buttons - UI

Week 6: Chat Flutter + Feedback

- Backend: RAG service: query embedding → vector search → top-3 chunks
- Backend: AI service: system prompt + RAG context + Gemini Flash
- Backend: Chat API with SSE streaming
- Flutter: Session list + new chat
- Flutter: Integrate With APIs

FEATURE 3: ASSESSMENT & SMART STUDY PLAN

Week 7: Study Plan System + Quiz

- Backend: StudyPlan model + distribution algorithm
- Backend: Auto-redistribute on missed days (cron job)
 - [GET /progress/summary](#) (syllabus %, quiz avg, streak)
- Flutter: Quiz screen + Results screen
- Flutter: Today's Mission card, Week strip, Progress rings
- Flutter: integrate with APIs (Quiz)

Week 8: Flashcards + Progress

- Backend: Flashcard model + auto-generation per lesson
- Backend: Spaced repetition logic (Leitner system)
- Flutter: AI recommendations (weak topics, due lessons)
- Push notification setup (FCM)

MONTH 3

FEATURE 4: INTERACTIVE LAB & TOOLS

Week 9: Periodic Table + Equation Balancer

- Periodic table JSON data (118 elements, Arabic names)
 - [GET /lab/elements](#), [POST /lab/balance](#)
 - Flutter: Lab home grid (4 tool cards from prototype)
 - Flutter: Periodic table (color-coded) + element detail sheet
 - Flutter: Equation balancer UI + step-by-step
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QA + POLISH + RELEASE

FEATURE 5: EXPERIENCE POINTS, PROFILE & POLISH

Week 10: XP(Experience Points) + Achievements

- Backend: XP(Experience Points) system (50/lesson, 30/quiz, 20/streak)
- Level calculation (Beginner → Scholar → Expert → Master)
- Achievement model + unlock conditions
- Flutter: Profile screen (avatar, stats grid, badges, settings)
- Flutter: Home hero (greeting, streak badge, XP badge)

QA PHASE

Week 11: Alpha Testing

- Security audit (JWT, rate limiting, input sanitization)
- Stress testing (100+ concurrent users)
- Cross-device testing (Android + iOS, various sizes)
- Memory leak testing

Week 12: Beta + Release

- Beta with 9 Grade 9 students (3-day usage)
- Fix top UX friction points
- Full regression (all QA checkboxes)
- App Store + Google Play submission

If we Have Time: Offline + Bilingual + Voice

- Hive local caching (last 10 lessons + flashcards)
- Offline banner + sync on reconnect
- Quick action grid on Home